paradigm in his 1962 book *The Structure of Scientific Revolutions*—one of the most insightful books I have ever read.

A paradigm is a complete set of tools that a community of scientists uses in its research on some topic, and in communicating the results of its research to others. The curved spacetime viewpoint on general relativity is one paradigm; the flat spacetime viewpoint is another. Each of these paradigms includes three basic elements: a set of mathematically formulated *laws* of physics; a set of *pictures* (mental pictures, verbal pictures, drawings on paper) which give us insight into the laws and help us communicate with each other; and a set of *exemplars*—past calculations and solved problems, either in textbooks or in published scientific articles, which the community of relativity experts agrees were correctly done and were interesting, and which we use as patterns for our future calculations.

The *curved spacetime paradigm* is based on three sets of mathematically formulated laws: Einstein's field equation, which describes how matter generates the curvature of spacetime; the laws which tell us that perfect rulers and perfect clocks measure the lengths and the times of Einstein's curved spacetime; and the laws which tell us how matter and fields move through curved spacetime, for example, that freely moving bodies travel along straight lines (geodesics). *The flat spacetime paradigm* is also based on three sets of laws: a law describing how matter, in flat spacetime, generates the gravitational field; laws describing how that field controls the shrinkage of perfect rulers and the dilation of the ticking rates of perfect clocks; and laws describing how the gravitational field also controls the motions of particles and fields through flat spacetime.

The pictures in the curved spacetime paradigm include the embedding diagrams drawn in this book (for example, the left half of Figure 11.1) and the verbal descriptions of spacetime curvature around black holes (for example, the "tornado-like swirl of space around a spinning black hole"). The pictures in the flat spacetime paradigm include the right half of Figure 11.1, with the ruler that shrinks when it turns from circumferential orientation to radial, and the verbal description of "a gravitational field controlling the shrinkage of rulers."

The exemplars of the curved spacetime paradigm include the calculation, found in most relativity textbooks, by which one derives Schwarzschild's solution to the Einstein field equation, and the calculations by which Israel, Carter, Hawking, and others deduced that a black hole has